MEDIUM ALGEBRA

Linear inequalities in one or two variables

02

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

$y \le 3x + 1$
$x - y > 1$

Which of the following ordered pairs (x, y) satisfies the system of inequalities above?

A. $(-2, -1)$

B. $(-1, 3)$

C. $(1, 5)$

D. $(2, -1)$

Q2 ALGEBRA

$$y \leq x$$
$$y \leq -x$$

Which of the following ordered pairs (x, y) is a solution to the system of inequalities above?

A. $(1, 0)$

B. $(-1, 0)$

C. $(0, 1)$

D. $(0, -1)$

Q3 ALGEBRA

For a snowstorm in a certain town, the minimum rate of snowfall recorded was 0.6 inches per hour, and the maximum rate of snowfall recorded was 1.8 inches per hour. Which inequality is true for all values of s, where s represents a rate of snowfall, in inches per hour, recorded for this snowstorm?

A. $s \geq 2.4$

B. $s \geq 1.8$

C. $0 \leq s \leq 0.6$

D. $0.6 \leq s \leq 1.8$

Q4

ALGEBRA

$$H = 120p + 60$$

The Karvonen formula above shows the relationship between Alice’s target heart rate H, in beats per minute (bpm), and the intensity level p of different activities. When $p = 0$, Alice has a resting heart rate. When $p = 1$, Alice has her maximum heart rate. It is recommended that p be between 0.5 and 0.85 for Alice when she trains. Which of the following inequalities describes Alice’s target training heart rate?

A. $120 \leq H \leq 162$

B. $102 \leq H \leq 120$

C. $60 \leq H \leq 162$

D. $60 \leq H \leq 102$

Q5

GRID-IN ALGEBRA

An event planner is planning a party. It costs the event planner a onetime fee of $\$35$ to rent the venue and $\$10.25$ per attendee. The event planner has a budget of $\$300$. What is the greatest number of attendees possible without exceeding the budget?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ALGEBRA

$$y < -4x + 4$$

Which point (x, y) is a solution to the given inequality in the xy-plane?

A. (-4, 0)

B. (0, 5)

C. (2, 1)

D. (2, -1)

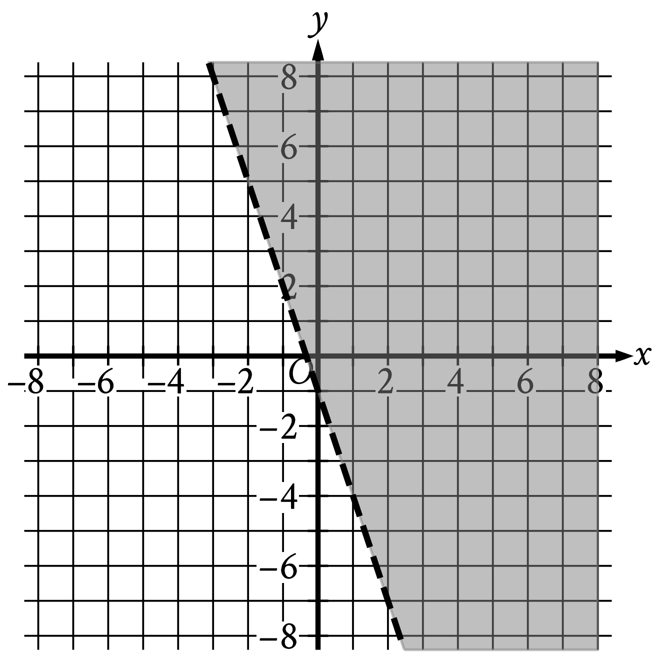
Q7

GRID-IN ALGEBRA

A number x is at most 17 less than 5 times the value of y. If the value of y is 3, what is the greatest possible value of x?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The boundary of the inequality is a dashed line.
    - The line slants sharply down from left to right.
    - The line passes through the following points:
        - (negative 1 comma 2)
        - (0 comma negative 1)
- The area above and to the right of the boundary is shaded.

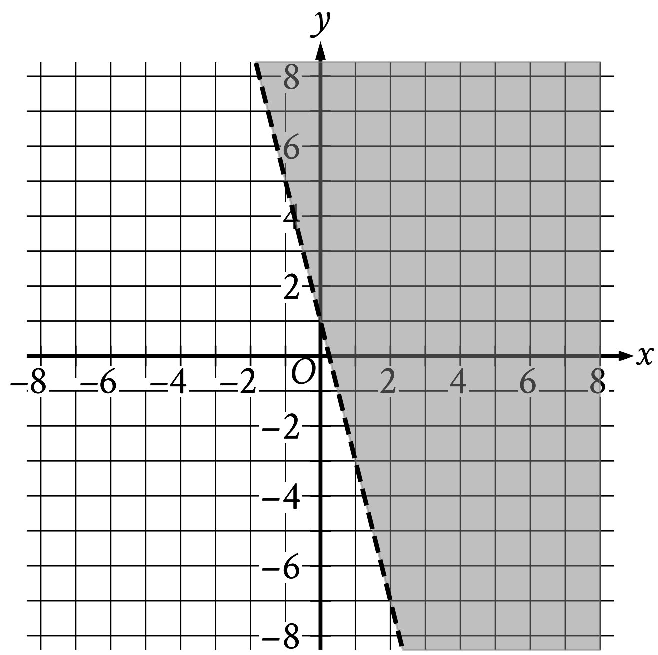
The shaded region shown represents the solutions to which inequality?

A. $y < -1 + 3x$

B. $y < -1 - 3x$

C. $y > -1 + 3x$

D. $y > -1 - 3x$



- The boundary of the inequality is a dashed line.
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 5)
    - (0 comma 1)
    - (1 comma negative 3)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y < 1 + 4x$

B. $y < 1 - 4x$

C. $y > 1 + 4x$

D. $y > 1 - 4x$

Q10 ALGEBRA

$$y > 4x + 8$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
2	19
4	30
6	41

B.

x	y
2	8
4	16
6	24

C.

x	y
2	13
4	18
6	23

D.

x	y
2	13
4	21
6	29